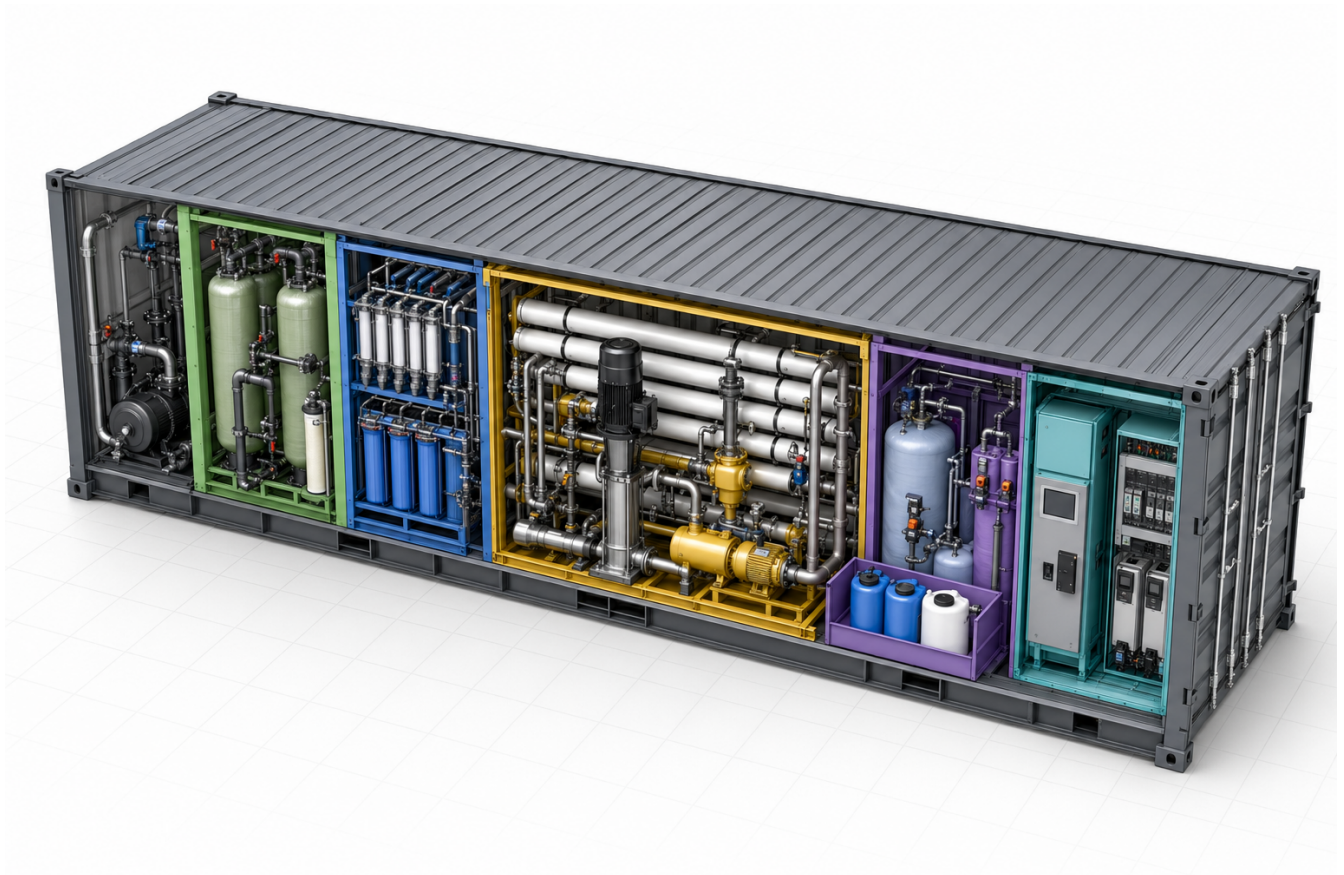


Containerised desalination — 500 m³/day SWRO

Revision A · Fractional Forge · In build

Forge project pack

Project name	Containerised desalination — 500 m ³ /day SWRO
Revision	Rev A
Project created	25 Apr 2026, 04:42
Brief locked	25 Apr 2026, 06:09
Shipped	Not shipped yet
Document generated	25 Apr 2026, 06:18



Illustrative only — not a technical specification. All renders in this document are generated for visual reference; component arrangement, proportions, and identities may differ from the final engineered assembly.

TOTALS AT A GLANCE

MODULES 7	KEY PARTS 46	BOM ROWS 68	SUPPLIERS 32
FAILURE MODES 28	OPEN QUESTIONS 21	STANDARDS 8	REVIEWS 0
UNIT COST £247,980	CEILING £350,000	HEADROOM £102,020	

Contents

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04	4. BOM master
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1. Brief

FOUNDER SUBJECT

A 40-foot containerised seawater reverse-osmosis desalination unit producing 500 cubic metres per day of potable water from seawater feed. Pre-treatment, two-pass reverse osmosis, post-treatment (remineralisation and chlorination), 50 per cent recovery rate, energy-recovery device on the brine stream. Target specific energy consumption under 3.5 kilowatt-hours per cubic metre. Designed for coastal disaster relief, remote-island municipal water, and construction-site freshwater supply. Target installed capital cost under 350,000 pounds. Must ship complete on a flatbed lorry, plug into a three-phase mains supply or local diesel genset, and start producing within four hours of arrival on site.

MISSION

To deliver field-deployable, containerised seawater desalination that produces potable water within four hours of site arrival, enabling rapid water security in coastal disaster, remote-island, and construction contexts.

USE CASE

A single 40-foot ISO container is delivered by flatbed lorry, connected to three-phase mains or a diesel genset, and begins producing 500 cubic metres per day of potable water within four hours of arrival.

TARGET CUSTOMERS

Humanitarian relief organisations, remote-island municipal water authorities, construction-site operators

WHY NOW

Accelerating frequency of coastal disasters, growing UN and government procurement for rapidly-deployable WASH infrastructure, and the commercial maturation of isobaric energy-recovery devices that bring seawater RO specific energy below 3.5 kWh/m³ at containerised scale — making self-contained units economically viable against trucked or bottled water.

Constraints declared

Unit cost ceiling	£350,000
Max mass	17500.00 kg
Target process	not declared
Target material	not declared
Tolerance target	not declared
Quantity target	not declared

2. Regulatory posture

NSF/ANSI 61

not-started

Drinking water system components — health effects

Mandatory certification for all wetted components in the product-water path, including gaskets, pipe, vessel liners, and post-treatment media.

NSF/ANSI 372

not-started

Drinking water system components — lead content

Requires weighted average lead content of wetted surfaces on fittings and valves to be 0.25% or below.

ISO 668

not-started

Series 1 freight containers — classification, dimensions and ratings

Governs the external envelope and gross mass of the 40-foot ISO container.

ISO 1496-1

not-started

Series 1 freight containers — specification and testing: general cargo containers

Defines structural requirements for the container shell, corrugated walls, and floor.

ISO 1161

not-started

Series 1 freight containers — corner fittings

Specifies corner casting geometry used for lifting and lashing the unit.

ISO 24516-4

not-started

Guidelines for the management of assets of water and wastewater systems — water treatment plants

Asset management and lifecycle reference framework applicable to the operational phase of the plant.

EN 12255

not-started

Wastewater treatment plants — general construction principles

Referenced for general design principles covering safety, automatic control, and disinfection provisions; Parts 10, 12, and 14 are most pertinent.

WHO Guidelines for Drinking-water Quality

not-started

WHO drinking water quality guidelines

Defines the potable water quality targets that post-treatment remineralisation and chlorination must meet.

3.1 · ISO Container Shell & Skid Frame

Module id: container_shell

4200.00 kg · 10 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Provides the 40ft ISO 1496 transportable enclosure and internal structural skid that mounts and protects all process equipment.

WHY IT MATTERS

This module defines the entire product's transportability, deployment speed, and structural integrity. Without ISO-compliant lift points and a dimensionally correct envelope it cannot ship by flatbed or crane, defeating the core value proposition of containerised deployment. The skid enables factory pre-assembly and FAT, which is essential to hit the 4-hour on-site startup target. It also concentrates the ~17.5 t wet operating load into a defined footprint so site foundation requirements can be specified, and it provides a corrosion-resistant, weatherproof, thermally insulated environment that protects ~£250k of process equipment in coastal and marine atmospheres for the 15–25 year design life.

DESCRIPTION

The container shell and skid frame is the primary structural and environmental envelope for the entire desalination plant. The outer enclosure is a standard 40ft ISO 1496-1 Corten weathering-steel shipping container (12,192 × 2,438 × 2,591 mm) modified with a personnel side-door, ventilation louvres, an extract fan, and cable/pipe penetration glands sealed with EPDM grommets. The container retains its ISO 1161 corner castings so it can be lifted by standard spreader, twistlocked to a flatbed lorry, and stacked during shipping. Internally a welded structural skid (~11,000 × 2,300 mm) carries every process module, decoupling equipment loads from the relatively flexible container floor and allowing complete factory pre-assembly and FAT before shipment. The skid is fabricated from SS304 (or epoxy-coated carbon steel for cost-down variants) RHS sections, sized to leave ~70 mm clearance to each container side-wall for cable trays and low-pressure piping, plus a 1,500 mm clear pull-space at one end for 8040 RO element replacement.

KEY PARTS (7)

- 40ft ISO 1496-1 Corten steel container, 12,192 × 2,438 × 2,591 mm external, 1.6–2.0 mm corrugated wall panels, ISO 1161 corner castings at all 8 corners, marine plywood floor on steel cross-members
- Internal process skid frame, ~11,000 × 2,300 mm footprint, welded SS304 RHS 100×100×6 mm sections (or carbon steel with marine epoxy coating), bolted to container floor cross-members via M16 A4 stainless fasteners
- Vibration isolation mounts (8 off) between HP pump baseplate and skid, rubber-in-shear type, 50–100 Hz isolation, 200 kg load rating each
- Side-wall personnel door (lockable, 800 × 1,900 mm) and end-wall double doors (standard ISO) with retained 1,500 mm membrane pull-space at one end
- Forced-ventilation louvres with insect mesh (2 off, 400 × 400 mm) and roof-mounted extract fan (IP55, 500 m³/h) for internal heat dissipation
- Internal 50 mm rockwool thermal/acoustic insulation lining on walls and roof, sheathed in 0.7 mm GRP liner panels
- Lifting/lashing hardware: 4× ISO 1161 corner castings rated 30 t vertical, twistlock-compatible for flatbed and crane handling

FAILURE MODES (4)

- Galvanic and chloride-induced corrosion of carbon steel skid members in coastal salt-spray environments leading to structural weakening over 10+ years
- Container floor cross-member deflection or fatigue cracking under concentrated HP pump dynamic loads if vibration isolation is inadequate
-

Internal condensation and humidity build-up corroding electrical enclosures and pipe supports if ventilation is undersized for tropical deployment

- Corner-casting weld failure or distortion from repeated lift/lash cycles if container is structurally compromised by door/louvre cut-outs without proper reinforcement

OPEN QUESTIONS (3)

- Whether standard-height (2,591 mm) container provides sufficient internal headroom for 8" pressure vessel manifolding and overhead cable trays, or if High-Cube (2,896 mm) variant is required
- Required floor cross-member reinforcement under HP pump and ERD locations to handle dynamic loads — needs FEA or supplier baseplate spec
- Site foundation specification (concrete pad thickness, levelness tolerance) required to support 17.5 t wet weight and meet 4-hour startup commitment

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
40ft ISO container shell — Standard shipping container, used condition.	make	Manual/Assembly · Other	£3,500
Internal skid frame (SS304 RHS) — Welded SS304 RHS frame, fabricated to size.	make	Welding · Stainless 304	£1,800
Vibration isolation mounts (×8) — Rubber-in-shear type, 200 kg capacity each.	make	Manual/Assembly · Other	£520
Side door & double doors — Standard ISO container doors, lockable.	buy	—	£1,200
Forced ventilation louvres with mesh (×2) — 400×400mm louvre panels with insect mesh.	buy	—	£180
Roof extract fan, IP55, 500 m3/h — Standard industrial extract fan.	buy	—	£250
Rockwool insulation (50mm) — 40 sqm needed, standard thermal/acoustic.	buy	—	£400
GRP liner panels (0.7mm) — Sheathing for walls/ceiling, ~40 sqm.	make	Manual/Assembly · Fibreglass	£350
M16 A4 stainless fasteners (×40) — Bolts for skid to floor, standard hardware.	buy	—	£60
Corner castings & lifting hardware — ISO 1161 castings, twistlocks included.	make	Casting · Other	£450
Labour	£800 — 2 workers × 10 hrs: cut/weld skid, fit insulation, install parts.		
Per-unit total	£9,510		

ASSUMPTIONS

- Container is used, estimated £3500.
- Labour rate £40/hr, 20 man-hours.

Majority buy items except custom welded skid frame.

3.2 · Pre-treatment Module

Module id: pretreatment

850.00 kg · 8 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Filters raw seawater through multimedia/UF and 5µm cartridge filters to protect downstream RO membranes from particulates and fouling.

WHY IT MATTERS

Pre-treatment is the single largest determinant of RO membrane life and operating cost. Inadequate filtration drives the silt density index (SDI) above 5, causing rapid colloidal fouling, increased trans-membrane pressure, higher specific energy consumption, and premature membrane replacement (membranes are ~20–25% of CAPEX). Reliable pre-treatment is also essential to the 4-hour startup target, since the unit must reach stable SDI<3 quickly on commissioning.

DESCRIPTION

The pre-treatment module is a multi-stage filtration train mounted at the feed end of the internal skid, occupying approximately 2,500 mm of skid length along the container side-wall. Raw seawater enters via a DN80–DN125 flanged intake, passes through a low-pressure booster pump, then through a multimedia filter (graded silica sand, anthracite, and garnet over a gravel support) which removes suspended solids >10 µm; an ultrafiltration skid may substitute or supplement the multimedia stage where feedwater turbidity or biological loading is high. Downstream, antiscalant and sodium bisulphite (for residual chlorine reduction) are dosed inline before a final 5 µm polypropylene cartridge filter polishes the stream and protects the high-pressure pump and RO membranes.

KEY PARTS (6)

- Multimedia filter vessel: Ø 900 mm × 2,200 mm FRP pressure vessel with graded sand/anthracite/garnet bed, design pressure 6 bar, top-mount multiport valve for backwash sequencing
- Ultrafiltration module (optional/alternative): pressurised hollow-fibre PVDF UF skid, 0.02 µm nominal pore size, 2 × modules at ~25 m³/h capacity each, integrated air-scour and CEB ports
- 5 µm cartridge filter housing: 316L stainless steel multi-cartridge housing, 7-round 40" polypropylene melt-blown cartridges, DN80 PN16 flanged inlet/outlet, differential pressure transmitter across housing
- Low-pressure feed booster pump: end-suction centrifugal, 25 m³/h @ 4 bar, 5.5 kW IE3 motor, duplex stainless steel wetted parts
- Antiscalant and sodium bisulphite dosing skids: 2 × diaphragm metering pumps with 60 L HDPE day tanks and calibration columns
- Instrumentation: feed pressure (PT), differential pressure (DPT) across each filter stage, SDI sample point, turbidity meter, ORP probe downstream of bisulphite injection

FAILURE MODES (4)

- Cartridge filter blinding from algal bloom or sediment surge causing high differential pressure and HP pump cavitation
- Multimedia filter channelling or media loss through underdrain failure, allowing particulates to bypass and foul RO membranes
- Bisulphite dosing pump failure leading to chlorine breakthrough and oxidative destruction of polyamide RO membranes
- Biofouling of UF fibres during shutdown periods if preservation/flush protocol is not followed

OPEN QUESTIONS (3)

•

Whether multimedia or UF is the baseline configuration — depends on target deployment feedwater quality envelope (open-intake vs beachwell)

- Backwash water source and discharge routing — self-supplied from permeate tank or external utility connection
- Required SDI guarantee at module outlet (typically <3) and corresponding cartridge change-out interval for design feed conditions

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Multimedia filter vessel, FRP, 900×2200mm — Standard FRP vessel with multiport valve.	buy	—	£2,800
Filter media (sand/anthracite/garnet) — Graded media for 6 bar vessel.	buy	—	£600
Ultrafiltration module skid (2× modules) — PVDF UF, 25 m3/h each, includes air-scour.	buy	—	£8,500
5µm cartridge filter housing, 316L, 7×40" — Flanged housing with differential pressure tx.	make	Manual/Assembly · Stainless Steel	£1,800
Polypropylene melt-blown cartridges (7) — Standard 40" cartridges, £20 each.	buy	—	£140
Feed booster pump, 25 m3/h @ 4 bar — Centrifugal, duplex SS, 5.5 kW IE3.	buy	—	£2,200
Antiscalant dosing skid — Diaphragm pump + 60L HDPE tank.	buy	—	£900
Sodium bisulphite dosing skid — Identical to antiscalant skid, separate unit.	buy	—	£900
Instrumentation set (PT, DPT, turbidity, ORP) — Includes SDI sampler, all transmitters.	buy	—	£1,600
Mounting frame for UF/MMF — Welded mild steel frame, epoxy coated.	make	Welding · Mild Steel	£350
Pipework and fittings (SS304) — Interconnecting pipes, valves, unions.	buy	—	£500
Labour	£600 — 2 workers × 7.5 hrs: mount vessels, connect piping, wire.		
Per-unit total	£20,390		

ASSUMPTIONS

- UF module is a typical PVDF skid, not custom.
- Labour includes skid assembly and plumbing.

Mostly COTS equipment, minimal custom fabrication.

3.3 · High-Pressure Pump & Energy Recovery Device

Module id: hp_pump_erd

850.00 kg · 14 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Pressurises filtered feed to 55-70 bar for RO and recovers pressure energy from the brine reject stream via an isobaric pressure exchanger.

WHY IT MATTERS

Without the ERD, SWRO specific energy consumption would be 6-8 kWh/m³, doubling running costs and violating the <3.5 kWh/m³ target. The HP pump alone consumes ~70-80% of the unit's electrical load, so its efficiency, reliability and correct matching to the ERD directly govern OPEX, fuel consumption when running on diesel genset, and overall commercial viability of the product.

DESCRIPTION

This module is the hydraulic and energetic heart of the desalination unit. The high-pressure pump (typically a Danfoss APP axial-piston or Grundfos BMS-class multistage, 75 kW nominal) draws filtered seawater at low pressure and delivers it at 55-70 bar to the first-pass RO train. In parallel, an isobaric rotary pressure exchanger (Danfoss iSave or ERI PX) accepts the high-pressure brine reject (~55-65 bar) leaving the membranes and transfers ~96-98% of its pressure energy directly into a portion of the incoming feed stream, which is then trimmed up to RO pressure by a small booster pump. Only the net permeate-flow equivalent is pressurised by the main HP pump, which is what reduces specific energy consumption to below 3.5 kWh/m³.

Wetted parts use super-duplex stainless steel and engineered ceramics (alumina rotor in the ERD) to resist chloride pitting and stress-corrosion cracking at SWRO conditions. The pump and ERD are mounted on the carbon-steel skid via anti-vibration isolators and connected with Duplex 2205 Sch10 piping rated PN100, with PED-certified relief valves, isolation ball valves, and a low-pressure suction strainer. The motor is a TEFC IE3 IEC frame driven through a VFD in the MCC for soft-start and flow trim.

KEY PARTS (6)

- High-pressure feed pump: Danfoss APP 21-series axial-piston or Grundfos BMS hp pump, 75 kW IEC frame motor, 55-70 bar discharge, ~21 m³/h flow, duplex SS / super-duplex wetted parts
- Booster pump: Grundfos CRN vertical multistage, 11-15 kW, raises filtered feed from ~2 bar to ~3-5 bar suction for HP pump and ERD high-pressure inlet
- Isobaric pressure exchanger ERD: Danfoss iSave 21 or ERI PX-Q140, ceramic rotor, ~96-98% pressure transfer efficiency, processing ~21 m³/h brine flow
- High-pressure piping spool set: Duplex 2205 Sch10, DN50-DN80, PN100 flanges, including HP manifold, isolation ball valves and PRV
- Pressure relief valve: spring-loaded SS316L, set 75 bar, DN25, certified to PED 2014/68/EU
- Instrumentation: HP discharge pressure transmitter (0-100 bar), feed/brine flow meters (electromagnetic DN65), motor RTDs and vibration sensor on pump bearing

FAILURE MODES (4)

- Chloride stress-corrosion cracking of high-pressure piping or fittings if non-duplex stainless steel is inadvertently used in wetted path
-

ERD ceramic rotor seizure or accelerated wear from particulate carryover (>5 µm) due to upstream cartridge filter bypass or rupture

- HP pump cavitation and bearing failure from insufficient suction pressure if booster pump trips or pre-treatment differential pressure rises
- Motor inrush trip on diesel genset start due to inadequate VFD ramp configuration or undersized generator

OPEN QUESTIONS (3)

- Final selection between Danfoss APP and Grundfos BMS hp pump pending efficiency curve and lead-time comparison at 21 m³/h duty point
- Exact ERD model sizing (iSave 21 vs iSave 40 vs PX-Q140) pending confirmed recovery rate (35% vs 50%) and brine flow
- Pulsation damper requirement on HP discharge — depends on pump type (piston pumps typically need accumulator, multistage do not)

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
High-pressure feed pump, 75 kW, 70 bar — Danfoss APP21 or similar, super-duplex wetted parts.	buy	—	£18,000
IEC motor, 75 kW — Standard frame size, IE3 efficiency.	buy	—	£3,500
Booster pump, CRN 11-15 kW — Vertical multistage Grundfos, raises feed pressure.	buy	—	£3,500
Isobaric ERD, iSave21 or PX-Q140 — High efficiency ceramic rotor, ~21 m³/h.	buy	—	£14,000
High-pressure piping spool set, Duplex 2205 — DN50-DN80, PN100 flanges, pre-fabricated set.	make	Manual/Assembly · Other	£2,500
Pressure relief valve, SS316L, 75 bar — Certified to PED, DN25.	buy	—	£400
Isolation ball valves (×4), Duplex — DN50-DN80, high pressure rated.	buy	—	£600
Instrumentation: pressure transmitter, flow meters, RTDs — 0-100 bar, DN65 mag meters, motor sensors.	buy	—	£2,500
Baseplate for pump & ERD — Welded SS304 frame, machined mounting surfaces.	buy	—	£600
HP manifold block, Duplex — CNC machined Duplex 2205 block, high pressure.	make	CNC Machining · Duplex 2205	£1,200
Labour	£800 — 2 workers × 10 hrs: assemble, pipe, align pump/ERD.		
Per-unit total	£47,600		

ASSUMPTIONS

- HP pump and ERD are major cost items, market verified.
- Labour includes alignment and pressure testing.

Primary costs are high-value buy items; make parts are limited.

3.4 · Two-Pass RO Membrane Train

Module id: ro_train

1800.00 kg · 10 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Separates feed into permeate and brine using 8040 membranes in FRP pressure vessels arranged in two passes for potable-quality output.

WHY IT MATTERS

This is the heart of the desalination process — it determines product water quality, recovery rate, energy consumption, and membrane replacement cost (the largest single OPEX item). Membrane selection and array configuration directly drive whether the unit hits its <3.5 kWh/m³ SEC target and meets WHO potable boron/chloride limits.

DESCRIPTION

The RO train is the core separation stage: a horizontally-mounted bank of 8" filament-wound FRP pressure vessels arranged in two passes. The first pass is a single-stage SWRO array (typically 6-7 vessels × 6 elements = 36-42 membranes) operating at 55-70 bar to reject ~99.8% of dissolved salts at 50% recovery. The second pass treats the first-pass permeate through 1-2 BWRO vessels at 10-15 bar to polish boron and residual chlorides to WHO potable limits. Vessels are end-port or side-port FRP rated to 1000 psi, mounted axially along the container length on HDPE-lined steel cradles to allow element extraction from the door end (e1.5 m pull space). Interconnect piping uses Duplex 2205 Sch10 on the high-pressure feed/concentrate side and SS316L on the permeate manifold; all wetted materials are NSF/ANSI 61 certified.

KEY PARTS (7)

- 1st Pass SWRO Membranes: 8040 spiral-wound thin-film composite (e.g. DuPont SW30HRLE-440i or Toray TM820V-440), 440 ft² active area, 99.8% nominal NaCl rejection, ~36-42 elements total
- 2nd Pass BWRO Membranes: 8040 brackish-water elements (e.g. DuPont BW30-400) for boron/chloride polishing, ~6-12 elements
- Pressure Vessels: 8" Ø FRP filament-wound, 1000 psi (69 bar) rated for 1st pass, 6-element configuration, side-port (e.g. Codeline 80S100-6 or Bel 8" 1000psi), ~6,100 mm length, qty 6-7 for 1st pass + 1-2 for 2nd pass
- Interconnect Piping: Duplex 2205 Sch10, DN50-DN80 high-pressure manifold with Victaulic grooved couplings, ASME B16.5 flanges
- Permeate Manifold: SS316L DN50, with conductivity probes at each vessel for permeate-quality monitoring
- Vessel Cradles: HDPE-lined steel saddles with stainless straps, mounted to skid frame on 1.5 m centres
- Instrumentation: 4-20 mA pressure transmitters (feed/concentrate/permeate), magnetic flowmeters on permeate and concentrate, conductivity sensors per pass

FAILURE MODES (4)

- Membrane fouling/scaling from inadequate pre-treatment causing pressure rise and flux decline
- FRP pressure vessel end-cap blow-out from improper retaining ring installation or fatigue cycling
- O-ring/brine-seal failure between elements causing internal bypass and salt passage spike
- Telescoping damage to membrane elements from pressure transients during startup/shutdown

OPEN QUESTIONS (3)

- Final element count and array configuration (6:1 vs 7:1 vessel:element) pending detailed projection (e.g. WAVE/IMSDesign) for site-specific TDS and temperature

- Whether second pass requires partial-flow (split-permeate) configuration to meet boron limits in warm tropical seawater
- Vessel manufacturer selection (Codeline vs Bel vs Protec) based on lead time and side-port vs end-port preference

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
SWRO Membranes (×36-42) — DuPont SW30HRLE-440i ~£1800/each at buy 42	—		£75,600
BWRO Membranes (×6-12) — DuPont BW30-400 ~£1200/each at 12	buy	—	£14,400
Pressure Vessels 1st Pass (×7) — FRP 6-element 1000psi Codeline ~£4000/each	buy	—	£28,000
Pressure Vessels 2nd Pass (×2) — Same type but fewer required	buy	—	£4,000
Interconnect Piping (Duplex 2205) — Welded Sch10 DN50-DN80 manifold make with flanges		Welding · Duplex 2205	£4,200
Permeate Manifold (SS316L) — Welded DN50 with ports for conductivity probes	make	Welding · SS316L	£1,500
Vessel Cradles (×9) — HDPE-lined saddles with straps, cut and bent	make	Sheet Metal · Mild Steel	£1,350
Pressure Transmitters 4-20mA (×3) — Industrial grade ~£300/each RS Components	buy	—	£900
Magnetic Flowmeters (×2) — DN50 magflow ~£1200/each e.g. Endress+Hauser	buy	—	£2,400
Conductivity Sensors (×2) — 4-electrode ~£300/each for permeate monitoring	buy	—	£600
Labour	£3,200 — 80 hrs skilled assembly/welding at £40/hr		
Per-unit total	£136,150		

ASSUMPTIONS

- 42 SWRO and 12 BWRO elements included
- Piping welded on-site; flanges are prefabricated buy parts

Major cost in membranes and vessels; piping and cradles fabricated.

3.5 · Post-treatment Module

Module id: posttreatment

650.00 kg · 6 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Remineralises permeate through a calcite contactor and chlorinates via NaOCl dosing to deliver potable water meeting WHO/EU standards.

WHY IT MATTERS

RO permeate is corrosive and unpalatable as-produced — without remineralisation it will leach metals from downstream pipework (failing NSF 61 / EU directive limits) and is rejected by consumers due to flat taste. Without chlorination, distribution-side biological recontamination is inevitable. This module is the legal gatekeeper between the technical RO process and a deliverable potable product; failure here invalidates the entire system's regulatory compliance regardless of upstream performance.

DESCRIPTION

The post-treatment module conditions low-TDS RO permeate (typically <300 mg/L, pH ~5.5-6.5, aggressive/corrosive) into stable, palatable potable water meeting WHO and EU Drinking Water Directive standards. It comprises three functional stages: (1) CO₂ injection to lower pH and create carbonic acid that dissolves calcium carbonate; (2) an upflow calcite contactor where permeate passes through a CaCO₃ media bed, raising calcium hardness to 40-80 mg/L as CaCO₃ and pH to 7.5-8.2 with positive Langelier Saturation Index; (3) sodium hypochlorite dosing (target 0.5-1.0 mg/L free chlorine residual) followed by a contact tank ensuring CT disinfection criteria. All wetted materials are NSF/ANSI 61 & 372 compliant: FRP vessel with food-grade liner, UPVC piping, EPDM/Viton gaskets, PVDF dosing wetted parts, 316L instrument wells. The module occupies approximately 2,000 × 1,500 mm of skid footprint at the discharge end of the container.

KEY PARTS (6)

- Calcite remineralisation contactor: GRP/FRP pressure vessel ~Ø600 × 1800 mm, NSF/ANSI 61 certified, filled with food-grade calcium carbonate media (CaCO₃, 1-3 mm granules, ~400 kg charge), top-mounted multiport valve for backwash, designed for 20-25 m³/h throughflow at 2-3 bar
- Sodium hypochlorite dosing skid: 60 L HDPE day tank with bund, 2× diaphragm metering pumps (e.g. Grundfos DDA 7.5-16, 0-7.5 L/h, duty/standby), PVC suction lance with low-level float, injection quill in DN50 PVC manifold, calibration column
- CO₂ dosing system (pH adjustment pre-calcite): 10 kg CO₂ cylinder with pressure regulator, solenoid valve, and PVDF injector — required to lower permeate pH and dissolve calcite (Langelier index correction)
- Inline instrumentation: pH sensor (Endress+Hauser Memosens CPS11D), free chlorine analyser (amperometric, 0-5 mg/L), conductivity probe, turbidity sensor, all 4-20 mA / Modbus to PLC
- Product water break tank: 1 m³ HDPE buffer tank with level switches, contact time chamber for chlorine disinfection (CT >15 mg·min/L)
- Post-treatment piping: DN50-DN80 UPVC PN10, EPDM gaskets (NSF 61), DN80 flanged outlet to external delivery

FAILURE MODES (4)

- Calcite media exhaustion/channelling causing under-dosed hardness, low pH product water, and corrosion of downstream metallic fittings
- NaOCl dosing pump failure or hypochlorite degradation (loses ~10% strength per month at 25°C) leading to insufficient chlorine residual and microbiological breakthrough
- CO₂ over-injection causing low pH and excessive calcium dissolution, or under-injection causing calcite bed to remain undissolved (no remineralisation)

- pH/chlorine sensor fouling or drift producing false telemetry, masking off-spec product water delivery

OPEN QUESTIONS (3)

- Final calcite consumption rate and refill interval — depends on permeate CO₂ dose and target hardness, affecting site-visit OPEX
- Whether single-stage calcite is sufficient or a downstream NaOH trim is needed to reach pH 8.0+ across full flow range
- Local regulatory variation: some jurisdictions mandate fluoridation or alternative disinfection (chloramine, UV) which would change the dosing skid configuration

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Calcite Contactor Vessel — GRP/FRP NSF certified ~600×1800 mm	buy	—	£3,500
Calcite Media (400 kg) — Food-grade 1-3mm granules ~£1.50/kg	buy	—	£600
Sodium Hypochlorite Day Tank (60L HDPE) — HDPE bunded day tank RSmake Components		Manual/Assembly · Polyethylene	£120
Diaphragm Metering Pumps (×2) — Grundfos DDA 7.5-16 duty/standby ~£1200 each	buy	—	£2,400
PVC Manifold and Injection Quill — Fabricated DN50 PVC assembly with quill	make	Cut and Bond · PVC	£180
CO ₂ Cylinder (10 kg) with Regulator — Standard CO ₂ cylinder and regulator	buy	—	£250
Solenoid Valve and PVDF Injector — PVDF injector and valve for CO ₂ dosing	buy	—	£150
pH Sensor (Memosens CPS11D) — Endress+Hauser ~£450 with cable	buy	—	£450
Free Chlorine Analyser (0-5 mg/L) — Amperometric ~£800 from suppliers	buy	—	£800
Conductivity Probe — Standard 4-20mA probe ~£250	buy	—	£250
Turbidity Sensor — Low range ~£350 for water quality	buy	—	£350
Product Water Break Tank (1 m ³ HDPE) — 1m ³ HDPE tank with level switches ~£800	make	Manual/Assembly · Polyethylene	£800
Post-treatment Piping UPVC DN50-DN80 — Cut and bonded UPVC with EPDM gaskets	make	Cut and Bond · UPVC	£400
Labour	£1,600 — 40 hrs plumbing assembly and installation		
Per-unit total	£11,850		

ASSUMPTIONS

- All sensors are buy items; piping fabricated on-site

Mostly off-shelf tanks and instruments; minimal fabrication of PVC piping.

3.6 · Clean-In-Place System

Module id: cip_system

180.00 kg · 6 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Stores and circulates membrane cleaning chemicals through the RO train to recover performance during scheduled maintenance cycles.

WHY IT MATTERS

Without periodic CIP, RO membranes foul progressively, causing permeate flux decline, increased differential pressure, higher specific energy consumption, and ultimately premature membrane replacement (membranes are ~20% of CAPEX). An integrated CIP system extends membrane life from ~3 years to 5-7 years, preserves the <3.5 kWh/m³ SEC target, and avoids the logistical cost of mobilising external cleaning rigs to remote disaster-relief or island sites.

DESCRIPTION

The Clean-In-Place (CIP) system is a self-contained chemical cleaning skid mounted at one end of the container, comprising a 500-1000 L polyethylene mixing tank, a 316L stainless-steel circulation pump (15-25 m³/h, 5.5-7.5 kW), a 5 µm cartridge guard filter, an immersion heater for warmed cleaning solutions, and manual valving to isolate and route flow through either the 1st-pass or 2nd-pass RO bank. Operating principle: during scheduled maintenance (typically every 3-12 months depending on fouling), the operator drains the tank, fills with RO permeate, manually doses citric acid (low-pH, for scale and metal foulants) or NaOH-based alkaline detergent (high-pH, for organic and biofouling), then activates circulation via the control panel. The pump recirculates the heated solution through the membranes at low pressure (3-5 bar) for 1-2 hours per cycle, followed by a soak and flush with permeate.

KEY PARTS (6)

- CIP Tank: 500-1000 L cylindrical polyethylene or GRP tank, vertical, with bolted lid, level switch, immersion heater (3-6 kW, 316L sheathed) for warm-water cleaning at 30-35°C, sized for one-pass flush of 1st-pass RO bank
- CIP Circulation Pump: 316L stainless steel horizontal end-suction or vertical multistage pump, 15-25 m³/h at 3-5 bar, 5.5-7.5 kW IE3 motor, mechanical seal compatible with citric acid (pH 2) and NaOH (pH 12)
- CIP Cartridge Filter: 5 µm polypropylene cartridge in 316L or reinforced PP housing, DN50 flanged, to prevent debris from re-entering membranes during cleaning
- Manual Diaphragm Valves & Hoses: PVC-U or 316L DN50-DN80 ball/diaphragm valves with EPDM seats, plus flexible reinforced EPDM hoses with camlock fittings for tying into RO feed/permeate manifolds
- Chemical Dosing Provision: Manual addition funnel with PP grating and eductor mixer for citric acid, sodium hydroxide, and proprietary alkaline/acidic detergents (e.g. Avista RoClean)
- Instrumentation: pH transmitter (0-14), conductivity sensor, temperature RTD (PT100), and pressure gauge on circulation loop, all wetted parts 316L or PVDF

FAILURE MODES (4)

- Chemical attack on pump seals or tank fittings due to incorrect chemical concentration or use of incompatible elastomers, causing leaks of corrosive fluid into the container
- Inadequate flush of residual cleaning chemicals before returning to service, contaminating product water and breaching NSF/ANSI 61 compliance
- Heater element scaling or dry-fire failure if level switch interlock fails, leading to tank damage or fire risk
- Cross-contamination between acid and alkaline cycles if tank is not properly rinsed, neutralising chemicals and rendering cleaning ineffective

OPEN QUESTIONS (3)

- Whether CIP tank volume should be sized for single-bank or whole-train cleaning, which affects tank footprint and water/chemical inventory
- Whether to specify automated dosing pumps and valve actuation (higher CAPEX, safer for unskilled operators) versus fully manual operation (lower cost, suits trained crew)
- Compatibility of selected EPDM/Viton elastomers with the full range of proprietary cleaning chemicals likely to be used across the unit's deployment scenarios

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
CIP Tank (500-1000L PE/GRP) — Vertical PE tank with lid ~£1200	make	Manual/Assembly · Fibreglass	£1,200
Immersion Heater (3-6kW 316L) — 316L sheathed heater ~£400	buy	—	£400
CIP Circulation Pump (316L) — Multistage 5.5-7.5kW with mechanical seal	buy	—	£3,500
Cartridge Filter Housing & Element (5µm) — 316L or PP housing with cartridge ~£300	make	Manual/Assembly · Other	£300
Manual Valves DN50-DN80 (×4) — PVC-U/316L ball/diaphragm with EPDM ~£200 each	buy	—	£800
Flexible Hoses with Camlocks (×2) — EPDM hose 2m with camlock fittings	buy	—	£300
Chemical Addition Funnel & Eductor — PP funnel with grating and mixer fabricated	make	Cut and Bond · PP	£80
pH Transmitter (0-14) — Industrial 4-20mA ~£300	buy	—	£300
Conductivity Sensor — For cleaning loop ~£200	buy	—	£200
Temperature RTD PT100 — PT100 probe ~£80	buy	—	£80
Pressure Gauge (316L/PVDF) — Glycerin filled ~£60	buy	—	£60
Labour	£1,200 — 30 hrs mechanical and plumbing installation		
Per-unit total	£8,420		

ASSUMPTIONS

- Pump and tank major costs; funnel fabricated from PP sheet

Majority buy items; small fabrication of chemical funnel.

3.7 · Control Panel & Motor Control Centre

Module id: control_mcc

250.00 kg · 10 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Houses PLC, HMI, VFDs and power distribution to sequence startup, monitor process variables, and manage 3-phase mains/genset power input.

WHY IT MATTERS

The MCC is the single point of integration for power, control, and safety: it determines whether the unit can genuinely start producing within four hours of arrival, whether it can ride through genset voltage dips, and whether unattended operation is safe. Soft-start/VFD control of the HP pump is essential to avoid tripping a typical site genset on inrush, and the PLC interlocks protect the £100k+ membrane investment from chemical or pressure excursions.

DESCRIPTION

The Control Panel & MCC is a wall-mounted IP65 stainless or coated-steel cubicle (~2.0 × 1.2 × 0.6 m) bolted to the interior side-wall of the container, housing all power distribution, motor control, and automation hardware for the desalination unit. Incoming 3-phase 380/415/460 V power passes through a main MCCB isolator, surge protection, and phase-monitoring relay before distributing to a 75 kW VFD driving the HP pump (the dominant load), DOL/soft-starters for auxiliary pumps, and a 24 VDC control bus feeding the PLC, HMI, and instrumentation. A Siemens or Allen-Bradley PLC executes the startup sequence (pre-treatment flush ' low-pressure fill ' HP ramp ' permeate-to-drain ' product-online), enforces interlocks (low feed pressure, high conductivity, high P), and modulates dosing/CIP commands.

KEY PARTS (8)

- Main MCC enclosure: IP65 painted mild steel or 316 stainless, ~2000 × 1200 × 600 mm, wall-mounted to container side-wall, with lockable double doors and internal gland plate; rated 100–160 A 3-phase 380/415/460 V 50/60 Hz incoming
- PLC + HMI: Siemens S7-1200/1500 or Allen-Bradley CompactLogix with 10" colour touchscreen HMI, Modbus TCP/Profinet to instruments, remote 4G/Ethernet gateway for telemetry
- Variable Frequency Drive for HP pump: 75 kW (sized for 45–95 kW pump range, select per pump) Danfoss VLT AQUA FC-202 or ABB ACS580, with EMC filter, dV/dt output reactor, and soft-start ramp to limit inrush below 2× FLC
- Incoming switchgear & protection: 250 A MCCB main isolator, surge protection device (Type 2, 40 kA), phase-failure/under-voltage relay, RCD 300 mA, and busbar distribution to MCC tiers
- Auxiliary contactors and motor starters: DOL/soft-start units for pre-treatment pumps, CIP pump, and dosing pumps (1.1–7.5 kW each), with thermal overloads and 24 VDC control circuits
- 24 VDC control power supply: 480 W redundant PSU (e.g. Phoenix Contact QUINT) with battery backup UPS for PLC/HMI ride-through (~15 min)
- I/O & instrumentation wiring: ~64 DI / 32 DO / 24 AI / 8 AO terminal blocks, 4–20 mA loops to pressure transmitters, conductivity sensors, flow meters, and ORP/pH probes
- Genset/mains changeover interface: ATS-ready terminal block, generator run/start dry contacts, and source-monitoring relay

FAILURE MODES (4)

- VFD overheating or derating in hot container interior (>45 °C ambient) leading to nuisance trips during summer deployments
- Ingress of salt-laden humid air into MCC causing corrosion of busbars and PCB contamination
- PLC I/O channel failure or sensor drift causing false interlock trips and unplanned shutdowns
- Inrush current on HP pump start tripping undersized site genset or mains breaker

OPEN QUESTIONS (3)

- Final HP pump motor rating (45–95 kW range) — drives VFD frame size and incoming breaker rating
- Whether customer requires SCADA/remote telemetry as standard or option, affecting gateway and cybersecurity scope
- Container internal ambient temperature profile — may require active MCC cooling (vortex cooler or AC unit) rather than passive ventilation

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Main MCC enclosure IP65 2000x1200x600mm painted mild steel — Custom welded enclosure, sheet metal, mild steel painted.	make	Sheet Metal · Mild Steel	£1,800
Siemens S7-1200 PLC + 10" HMI + 4G gateway — Standard industrial PLC/HMI package with telemetry.	buy	—	£3,500
Danfoss VLT AQUA FC-202 75kW VFD — Standard VFD for pump control.	buy	—	£3,200
250A MCCB main isolator — Standard MCCB, UK market price.	buy	—	£450
Surge protection device Type 2 40kA — Standard SPD, RS Components.	buy	—	£120
Phase failure relay — Standard protection relay.	buy	—	£80
RCD 300mA — Standard residual current device.	buy	—	£60
Busbar distribution kit — Standard busbar system for MCC.	buy	—	£200
DOL starters x4 (1.1-7.5kW) — Standard motor starters with overloads.	buy	—	£600
Soft starter x2 (1.1-7.5kW) — Standard soft starters for pumps.	buy	—	£500
24VDC redundant PSU 480W — Standard industrial power supply.	buy	—	£350
UPS battery backup 15min — Standard UPS for PLC ride-through.	buy	—	£250
Terminal blocks: 64 DI, 32 DO, 24 AI, 8 AO — Standard terminal blocks and loop wiring.	buy	—	£400
4-20mA loop cables and connectors — Standard instrumentation cables.	buy	—	£150
ATS changeover terminal block + relay — Standard ATS interface components.	buy	—	£150
Genset start/run dry contacts — Standard control relay.	buy	—	£50
Labour	£2,800 — 70 hours assembly, wiring, testing at £40/hr.		
Per-unit total	£14,060		

ASSUMPTIONS

- Standard industrial components used.
- Painted mild steel enclosure, not stainless.
- No custom PLC programming included beyond hardware.
- Batch size 1, labour based on build complexity.

MCC built from custom enclosure plus standard switchgear, VFD, PLC, and control components, with moderate assembly labour.

4. BOM master (68 rows)

BOM derived from the module decomposition's keyParts, expanded into typed part rows. Part records live in the `parts` table and are joined back to modules via source_module_id.

PART #	NAME	MODULE	TYPE	PROCESS / MATERIAL	MASS	COST
CIP-005	CIP Hose & Valve Set — Reinforced PVC hose, manual ball valves, quick-connect couplings and DN25 piping set to route CIP fluid between tank, pump, filter, heater and RO train supply/return manifolds.	Clean-In-Place Sys-tem	make	other · PVC / Brass · $\pm\pm 2\text{mm}$	18.00kg	£245
CIP-ASY	Clean-In-Place System Assembly — Top-level Clean-In-Place system assembly integrating mixing tank, circulation pump, cartridge filter, immersion heater, and associated valves and hoses to circulate cleaning chemicals through the RO train.	Clean-In-Place Sys-tem	make	other · Mixed Assembly · $\pm\pm 2\text{mm}$	185.00kg	£4,850
CIP-PUR-001	CIP Mixing Tank — 500-litre vertical polyethylene mixing tank for preparing acid (citric) and alkaline (NaOH) cleaning solutions for the RO membranes. Includes lid, level indicator, and bottom drain.	Clean-In-Place Sys-tem	buy	—	28.00kg	£320
CIP-PUR-002	CIP Circulation Pump — Centrifugal CIP circulation pump, stainless steel wetted parts, sized for 15 m ³ /h at 3 bar to circulate cleaning chemicals through the RO train at recommended cross-flow velocity.	Clean-In-Place Sys-tem	buy	—	42.00kg	£1,450
CIP-PUR-003	CIP Cartridge Filter Housing — 20-inch double-cartridge filter housing (5µm) installed downstream of the CIP pump to capture particulates dislodged from membranes during cleaning cycles.	Clean-In-Place Sys-tem	buy	—	12.00kg	£285
CIP-PUR-004	CIP Immersion Heater — 9 kW flange-mounted electric immersion heater with thermostat, raises CIP solution to 35°C for effective membrane cleaning. Incolloy 825 sheath for chemical compatibility.	Clean-In-Place Sys-tem	buy	—	6.50kg	£420
CS-002	Internal Process Skid Frame — Welded structural skid frame fabricated from RHS sections to mount and locate all process equipment within the container, with lifting lugs and levelling pads. Transfers equipment loads to container floor and corner posts.	ISO Container Shell & Skid Frame	make	sheet_metal · Structural Steel · $\pm\pm 2\text{mm}$ flatness, $\pm 3\text{mm}$ overall	420.00kg	£2,200

CS-004	Side-Wall Personnel Door Assembly — Hinged personnel access door cut into the container side wall with reinforced frame, weather seal, panic bar, and lockset. Provides safe operator access without using the cargo end doors.	ISO Container Shell make & Skid Frame	sheet_metal · Corten / Mild Steel · ± 1.5 mm gap	55.00kg	£480
CS-007	Internal Insulation Lining Panel — Composite internal lining panel comprising rigid PIR insulation foam bonded to a GRP facing sheet, mechanically fixed to container interior walls and ceiling. Provides thermal insulation and a wipe-clean hygienic interior surface.	ISO Container Shell make & Skid Frame	sheet_metal · PIR Foam / GRP · ± 2 mm panel, ± 5 mm install	8.50kg	£78
CS-ASY	Container Shell & Skid Frame Assembly — Top-level assembly comprising the 40ft ISO container, internal process skid, vibration mounts, doors, ventilation, and insulation lining. Provides the structural and environmental envelope for the entire desalination plant.	ISO Container Shell make & Skid Frame	other · Mixed (Corten/Steel/Insulation) · ± 5 mm overall	4800.00kg	£14,500
CS-PUR-001	40ft ISO 1496-1 Corten Shipping Container — Standard 40ft High-Cube ISO 1496-1 certified Corten-A shipping container providing the transportable, weather-resistant enclosure with corner castings for crane lift and twist-lock securing on flatbed transport.	ISO Container Shell buy & Skid Frame	—	3800.00kg	£4,500
CS-PUR-003	Vibration Isolation Mount — Bonded rubber-to-metal anti-vibration mount installed between process skid and container floor to isolate pump/RO vibration from the container structure. Reduces transmitted noise and fatigue loading.	ISO Container Shell buy & Skid Frame	—	0.45kg	£13
CS-PUR-005	Ventilation Louvre with Insect Mesh — Weatherproof aluminium louvre with stainless insect mesh providing passive ventilation airflow into the container while excluding rain ingress and pests. Mounted in side wall opening.	ISO Container Shell buy & Skid Frame	—	4.50kg	£95
CS-PUR-006	Roof-Mounted Extract Fan — Roof-mounted axial extract fan with weather cowl providing forced ventilation to remove process heat and any chemical vapours from inside the container. IP55 rated for outdoor installation.	ISO Container Shell buy & Skid Frame	—	14.00kg	£285
CS-PUR-008	ISO 1161 Corner Casting — Standard corner casting for 40ft container, used at all 8 corners for stacking, lifting and twistlock securement during transport.	ISO Container Shell buy & Skid Frame	—	8.50kg	£28

CS-PUR-009	EPDM Cable/Pipe Penetration Gland — EPDM multi-diameter penetration gland for sealing cable and pipe entries through the container wall while maintaining IP65 weather/dust protection.	ISO Container Shell buy & Skid Frame	—	0.35kg	£23
HP-004	High-Pressure Piping Spool Set (Duplex 2205) — Set of machined, bent and welded high-pressure pipework spools in duplex 2205 connecting HP pump discharge, ERD ports and RO vessel feed/brine manifolds. Rated for 80 bar service.	High-Pressure Pump & Energy Recovery Device	make	machining · Duplex Stainless Steel · ± 1.0 mm length, $\pm 0.5^\circ$ angular, weld per ASME IX	110.00kg £5,800
HP-ASY	HP Pump & ERD Assembly — Top-level assembly bringing together the HP feed pump, booster pump, isobaric energy recovery device and interconnecting duplex pipework on a common sub-skid baseplate.	High-Pressure Pump & Energy Recovery Device	make	other · Mixed (assembly) · ± 2.0 mm assembly	520.00kg £0
HP-PUR-001	High-Pressure Feed Pump — High-pressure axial piston or multistage centrifugal feed pump delivering ~21 m ³ /h at 60-70 bar to drive the RO membrane train. Designed for continuous seawater duty with duplex/super duplex wetted parts.	High-Pressure Pump & Energy Recovery Device	buy	—	95.00kg £14,500
HP-PUR-002	Booster Pump (Multistage) — Multistage vertical centrifugal booster pump that lifts pre-treated feed from atmospheric to ~3-4 bar before the HP pump suction. Stainless construction sized for ~42 m ³ /h flow.	High-Pressure Pump & Energy Recovery Device	buy	—	75.00kg £4,200
HP-PUR-003	Isobaric Pressure Exchanger ERD — Isobaric pressure exchanger that recovers hydraulic energy from RO brine reject and transfers it to incoming feed, reducing HP pump load by ~60%. Ceramic rotor design with no electrical drive.	High-Pressure Pump & Energy Recovery Device	buy	—	65.00kg £18,500
HP-PUR-005	HP Pressure Relief Valve — High-pressure relief valve set to 75 bar, protecting the HP pump and RO train from overpressure events. Spring-loaded poppet design with manual reset and 1/2" BSP connections.	High-Pressure Pump & Energy Recovery Device	buy	—	1.80kg £285
MCC-001	MCC Enclosure — IP55 floor-standing sheet steel MCC enclosure (1800x600x2000mm) with lockable double doors, gland plates, and internal mounting frame for PLC, VFDs, breakers and terminal rails.	Control Panel & Motor Control Centre	make	sheet_metal · Mild Steel · ± 1.5 mm	145.00kg £1,850

MCC-ASY	Control Panel & MCC Assembly — Top-level control panel and motor control centre assembly housing PLC, HMI, VFDs, contactors and distribution to sequence the desalination process and manage power from mains or genset.	Control Panel & Mo-make tor Control Centre	other · Mixed Assembly · ±±2mm	220.00kg	£12,500
MCC-PUR-002	PLC Controller — Industrial PLC controller managing process sequencing, interlocks, and SCADA communications for the desalination skid. Modular rack-mount unit with Ethernet/IP and Modbus TCP support.	Control Panel & Mo-buy tor Control Centre	—	1.20kg	£1,850
MCC-PUR-003	HMI Touchscreen Panel — 10-inch colour TFT HMI touchscreen for operator interface, alarm display, and trend visualisation. IP65-rated front face for panel-door mounting.	Control Panel & Mo-buy tor Control Centre	—	1.80kg	£1,200
MCC-PUR-004	Variable Frequency Drive (75 kW) — 75 kW variable frequency drive for high-pressure RO feed pump motor, providing soft-start, speed control, and energy optimisation. Includes integrated EMC filter and DC choke.	Control Panel & Mo-buy tor Control Centre	—	42.00kg	£4,200
MCC-PUR-005	Main MCCB Isolator — Main moulded-case circuit breaker isolator for incoming 3-phase supply, providing overcurrent and short-circuit protection with lockable rotary handle. Sized for 250 A frame at 415 VAC.	Control Panel & Mo-buy tor Control Centre	—	4.50kg	£480
MCC-PUR-006	Surge Protection Device — Type 2 DIN-rail surge protection device protecting MCC and downstream electronics from transient over-voltages on the 3-phase + N supply. Plug-gable MOV modules with status indication.	Control Panel & Mo-buy tor Control Centre	—	0.50kg	£220
MCC-PUR-007	Motor Starter / Soft-Starter — Soft-starter for ancillary AC motors (booster pumps, dosing) providing controlled ramp-up/down to reduce inrush current and mechanical shock. Built-in bypass contactor and overload protection.	Control Panel & Mo-buy tor Control Centre	—	6.00kg	£650
MCC-PUR-008	24 VDC Power Supply with UPS — Industrial 24 VDC switch-mode power supply (480 W) with integrated UPS battery buffer module to ride through brief mains interruptions and ensure controlled shutdown. DIN-rail mounted.	Control Panel & Mo-buy tor Control Centre	—	3.20kg	£540
MCC-PUR-009	I/O Terminal Block Set — Set of DIN-rail spring-clamp terminal blocks for field I/O wiring, including feed-through, earth, and fused disconnect terminals with end plates and jumpers. Approx 200-terminal assortment.	Control Panel & Mo-buy tor Control Centre	—	2.50kg	£320

MCC-PUR-010	ATS Changeover Interface — Automatic Transfer Switch interface module for seamless changeover between three-phase mains supply and diesel genset. Provides phase-loss detection, source priority logic, and dry contacts to PLC.	Control Panel & Motor Control Centre	buy	—	8.50kg	£685
PO-007	Post-treatment Piping Spool Set (UPVC) — Set of fabricated UPVC piping spools, elbows, tees and flanged unions interconnecting the calcite contactor, dosing injection points, sensors and break tank within the post-treatment module.	Post-treatment Module	make	other · UPVC · ±±3mm on cut lengths, ±1° on flange faces	42.00kg	£680
PO-ASY	Post-treatment Module Assembly — Top-level post-treatment module assembly integrating remineralisation contactor, chlorination dosing, CO2 injection, instrumentation, break tank and interconnecting piping mounted on a sub-skid frame.	Post-treatment Module	make	other · Mixed (Skid: Carbon Steel, Vessels: HDPE/FRP, Piping: UPVC) · ±±5mm on skid, ±2mm on pipe interfaces	1850.00kg	£38,500
PO-PUR-001	Calcite Remineralisation Contactor Vessel — Pressurised calcite (CaCO3) contactor vessel through which RO permeate is passed to dissolve calcium carbonate, raising hardness, alkalinity and Langelier index for potable distribution.	Post-treatment Module	buy	—	320.00kg	£4,800
PO-PUR-002	Sodium Hypochlorite Dosing Skid — Packaged sodium hypochlorite (12% NaOCl) dosing skid with HDPE day tank, twin diaphragm metering pumps, calibration column and injection quill for residual chlorination of product water.	Post-treatment Module	buy	—	95.00kg	£3,200
PO-PUR-003	CO2 Dosing System — CO2 dosing system feeding gaseous CO2 upstream of the calcite contactor to lower permeate pH and accelerate calcium carbonate dissolution; includes regulator, mass flow controller, solenoid and injector.	Post-treatment Module	buy	—	45.00kg	£2,100
PO-PUR-004	pH Sensor — Industrial inline pH sensor with combination glass electrode and integrated Pt1000 temperature compensation, mounted in a 1" BSP tee for continuous monitoring of post-treatment water quality.	Post-treatment Module	buy	—	0.35kg	£420
PO-PUR-005	Free Chlorine Analyser — Amperometric free chlorine analyser with flow cell, measuring residual free chlorine 0-5 mg/L in product water to verify disinfection compliance; provides 4-20mA output to PLC.	Post-treatment Module	buy	—	2.80kg	£1,450

PO-PUR-006	Product Water Break Tank (HDPE) — Atmospheric HDPE break tank providing buffer storage of treated potable water with air gap for backflow prevention, level sensors and DN80 outlet flange to distribution pump.	Post-treatment Module	buy	—	65.00kg	£780
PT-008	Pre-treatment Pipework Spool Set — Set of machined and welded pipework spools connecting media filters, cartridge filters and dosing injection points within the pre-treatment module. Includes flanged connections and instrument tapplings.	Pre-treatment Module	make	machining · Super Duplex Stainless Steel · ±±1.0mm on length, ±0.5° angular	85.00kg	£3,200
PT-ASY	Pre-treatment Module Assembly — Complete pre-treatment skid assembly comprising multimedia filter, UF skid, cartridge filter and booster pump, pre-piped and skid-mounted for installation in the container.	Pre-treatment Module	make	other · Mixed (FRP, 316SS, PVC) · ±±2.0mm (assembly)	850.00kg	£32,000
PT-PUR-001	Multimedia Filter Vessel (FRP) — Vertical FRP pressure vessel filled with multi-media (anthracite/sand/garnet) for removal of suspended solids and turbidity from raw seawater feed; sized for ~25 m³/h service flow.	Pre-treatment Module	buy	—	180.00kg	£2,400
PT-PUR-002	Ultrafiltration Hollow-Fibre Skid — Pressurised hollow-fibre ultrafiltration skid (PVDF membranes, ~0.03µm) providing SDI<3 feed to the RO train; includes modules, manifolds and backwash valves.	Pre-treatment Module	buy	—	450.00kg	£18,500
PT-PUR-003	5µm Cartridge Filter Housing — Multi-round cartridge filter housing accommodating 5x 40" cartridges as final particulate guard (5µm) protecting the high-pressure pump and RO membranes.	Pre-treatment Module	buy	—	35.00kg	£950
PT-PUR-004	Cartridge Filter Element — 5 micron melt-blown polypropylene cartridge filter element, 40" length, DOE end caps; consumable item replaced per maintenance schedule.	Pre-treatment Module	buy	—	0.45kg	£9
PT-PUR-005	Low-Pressure Feed Booster Pump — Horizontal multistage centrifugal booster pump delivering ~25 m³/h at 4 bar to feed the pre-treatment train and overcome filter pressure drop.	Pre-treatment Module	buy	—	65.00kg	£2,850
PT-PUR-006	Antiscalant Dosing Skid — Packaged antiscalant dosing skid with metering pump, day tank, and injection quill to prevent scale formation on RO membranes. Includes calibration column and low-level switch.	Pre-treatment Module	buy	—	35.00kg	£1,850

PT-PUR-007	Sodium Bisulphite Dosing Skid — Sodium bisulphite dosing skid for dechlorination of feed water upstream of RO membranes. Includes diaphragm metering pump, 100L HDPE tank, agitator and injection assembly.	Pre-treatment Module	buy	—	38.00kg	£1,950
RO-004	RO Interconnect Piping Spool Set — Set of CNC-machined high-pressure interconnect piping spools linking pressure vessels, HP pump discharge, and ERD. Includes feed, concentrate, and crossover spools with grooved-end connections.	Two-Pass RO Membrane Train	make	machining · Super Duplex Stainless Steel · ±±0.2mm	28.00kg	£1,450
RO-005	Permeate Manifold — Low-pressure permeate collection manifold gathering product water from all pressure vessel end ports. CNC-machined header with multiple branch ports and sample/instrumentation tappings.	Two-Pass RO Membrane Train	make	machining · 316L Stainless Steel · ±±0.3mm	12.00kg	£580
RO-006	Pressure Vessel Cradle Saddle — Sheet-metal cradle saddle supporting 8" FRP pressure vessels on the skid frame with EPDM rubber liner to prevent FRP abrasion. Bolt-down design with slotted holes for thermal expansion.	Two-Pass RO Membrane Train	make	sheet_metal · 304 Stainless Steel · ±±0.5mm	3.20kg	£48
RO-ASY	Two-Pass RO Membrane Train Assembly — Complete two-pass reverse osmosis membrane train assembly comprising SWRO and BWRO pressure vessels, interconnect piping, manifolds, and saddle supports. Configured for 500 m³/day potable production.	Two-Pass RO Membrane Train	make	other · Mixed (FRP, 316SS, Composite) · ±±2.0mm assembly	850.00kg	£42,000
RO-PUR-001	1st Pass SWRO Membrane Element 8040 — First-pass seawater RO membrane element, 8" diameter x 40" length spiral-wound polyamide thin-film composite. Provides 99.8% NaCl rejection at 55-70 bar feed pressure.	Two-Pass RO Membrane Train	buy	—	15.00kg	£650
RO-PUR-002	2nd Pass BWRO Membrane Element 8040 — Second-pass brackish water RO membrane element, 8" x 40" spiral-wound polyamide TFC for polishing first-pass permeate to potable quality. Operates at 10-15 bar with high boron rejection.	Two-Pass RO Membrane Train	buy	—	14.00kg	£420
RO-PUR-003	8" FRP Pressure Vessel — 8-inch fiberglass-reinforced plastic pressure vessel rated for 80 bar service, housing up to 6 RO membrane elements in series. Side-port configuration with Victaulic feed/concentrate connections.	Two-Pass RO Membrane Train	buy	—	95.00kg	£1,850

SH-PUR-001	Pressure Transmitter (4-20 mA) — Loop-powered pressure transmitter for monitoring feed/filter pressures in pre-treatment lines. 4-20 mA output to PLC analog input with G1/4 process connection.	Pre-treatment Module	buy	—	0.06kg	£12
SH-PUR-002	Differential Pressure Transmitter — Differential pressure transmitter measuring pressure drop across multimedia and cartridge filters to indicate fouling and trigger backwash cycles. 4-20 mA HART output.	Pre-treatment Module	buy	—	3.20kg	£1,250
SH-PUR-003	Electromagnetic Flow Meter — Electromagnetic flow meter for measuring high-pressure feed flow to RO train with no moving parts and high accuracy on conductive seawater. DN50 flanged, 4-20 mA output.	High-Pressure Pump & Energy Recovery Device	buy	—	9.50kg	£1,850
SH-PUR-004	Conductivity Probe — Inline conductivity probe measuring permeate quality (µS/cm) to verify RO membrane performance and trigger divert-to-drain on out-of-spec output. 4-20 mA transmitter output.	Two-Pass RO Membrane Train	buy	—	0.35kg	£425
SH-PUR-005	Temperature RTD (PT100) — Class A PT100 RTD temperature sensor in 316L stainless thermowell for monitoring CIP fluid temperature during cleaning cycles. 4-wire connection to PLC.	Clean-In-Place System	buy	—	0.18kg	£65
SH-PUR-006	Diaphragm Metering Pump — Solenoid-driven diaphragm metering pump for accurate antiscalant and SMBS dosing into pre-treatment line. Adjustable stroke length and frequency, PVDF wetted parts for chemical compatibility.	Pre-treatment Module	buy	—	3.80kg	£685
SH-PUR-007	Stainless Steel Ball Valve — Manual full-bore stainless steel ball valve for CIP isolation and drain points. Lever handle, full-port for low pressure drop on cleaning fluid circulation.	Clean-In-Place System	buy	—	0.42kg	£28
SH-PUR-008	Victaulic Grooved Coupling — Mechanical grooved-end coupling for rapid joining of grooved stainless steel pipes in the RO train high-pressure manifold. Provides flexible connection allowing slight angular deflection and vibration absorption.	Two-Pass RO Membrane Train	buy	—	1.80kg	£45
SH-PUR-009	PN100 Flange (Duplex 2205) — PN100 weld-neck flange rated for 100 bar service on the high-pressure pump discharge and ERD ports. Duplex 2205 provides chloride pitting/crevice corrosion resistance for seawater RO.	High-Pressure Pump & Energy Recovery Device	buy	—	4.50kg	£95

SH-PUR-010	EPDM Gasket (NSF 61) — NSF/ANSI 61 certified EPDM elastomer gasket for potable water flange seals on post-treatment piping. Non-toxic and chlorine-resistant, suitable for chlorinated permeate service.	Post-treatment Module	buy	—	0.05kg	£5
SH-PUR-011	Stainless Steel Fastener Set (A4) — Assorted A4-80 (316) stainless steel bolt, nut and washer set for securing skid components, panels, and equipment within the ISO container. Marine-grade corrosion resistance for coastal/saline environment.	ISO Container Shell & Skid Frame	buy	—	8.00kg	£125

5. Cost waterfall

Cost estimates grounded against the material-properties library (47 rows, freshest 23 Mar 2026) and the `process\_capabilities` table (20 rows, freshest unknown). Costs sit in the project's ai_cost_estimates jsonb, keyed by module id.

UNIT COST (ALL-IN)	CEILING (BRIEF)	HEADROOM
£247,980	£350,000	£102,020

Per-module roll-up

MODULE	COST	% OF UNIT
ISO Container Shell & Skid Frame	£9,510	3.8%
Pre-treatment Module	£20,390	8.2%
High-Pressure Pump & Energy Recovery Device	£47,600	19.2%
Two-Pass RO Membrane Train	£136,150	54.9%
Post-treatment Module	£11,850	4.8%
Clean-In-Place System	£8,420	3.4%
Control Panel & Motor Control Centre	£14,060	5.7%

6. Risks register

Every failure mode and open question declared against each module, in one register. Until a dedicated risks store ships with severity + ownership, these rows are the canonical risk inventory.

ISO Container Shell & Skid Frame

KNOWN FAILURE MODES (4)

- Galvanic and chloride-induced corrosion of carbon steel skid members in coastal salt-spray environments leading to structural weakening over 10+ years
- Container floor cross-member deflection or fatigue cracking under concentrated HP pump dynamic loads if vibration isolation is inadequate
- Internal condensation and humidity build-up corroding electrical enclosures and pipe supports if ventilation is undersized for tropical deployment
- Corner-casting weld failure or distortion from repeated lift/lash cycles if container is structurally compromised by door/louvre cut-outs without proper reinforcement

OPEN QUESTIONS (3)

- Whether standard-height (2,591 mm) container provides sufficient internal headroom for 8" pressure vessel manifolding and overhead cable trays, or if High-Cube (2,896 mm) variant is required
- Required floor cross-member reinforcement under HP pump and ERD locations to handle dynamic loads — needs FEA or supplier baseplate spec
- Site foundation specification (concrete pad thickness, levelness tolerance) required to support 17.5 t wet weight and meet 4-hour startup commitment

Pre-treatment Module

KNOWN FAILURE MODES (4)

- Cartridge filter blinding from algal bloom or sediment surge causing high differential pressure and HP pump cavitation
- Multimedia filter channelling or media loss through underdrain failure, allowing particulates to bypass and foul RO membranes
- Bisulphite dosing pump failure leading to chlorine breakthrough and oxidative destruction of polyamide RO membranes
- Biofouling of UF fibres during shutdown periods if preservation/flush protocol is not followed

OPEN QUESTIONS (3)

- Whether multimedia or UF is the baseline configuration — depends on target deployment feedwater quality envelope (open-intake vs beachwell)
- Backwash water source and discharge routing — self-supplied from permeate tank or external utility connection
- Required SDI guarantee at module outlet (typically <3) and corresponding cartridge change-out interval for design feed conditions

High-Pressure Pump & Energy Recovery Device

KNOWN FAILURE MODES (4)

- Chloride stress-corrosion cracking of high-pressure piping or fittings if non-duplex stainless steel is inadvertently used in wetted path
- ERD ceramic rotor seizure or accelerated wear from particulate carryover ($>5\text{ }\mu\text{m}$) due to upstream cartridge filter bypass or rupture
- HP pump cavitation and bearing failure from insufficient suction pressure if booster pump trips or pre-treatment differential pressure rises
- Motor inrush trip on diesel genset start due to inadequate VFD ramp configuration or undersized generator

OPEN QUESTIONS (3)

- Final selection between Danfoss APP and Grundfos BMS hp pump pending efficiency curve and lead-time comparison at $21\text{ m}^3/\text{h}$ duty point
- Exact ERD model sizing (iSave 21 vs iSave 40 vs PX-Q140) pending confirmed recovery rate (35% vs 50%) and brine flow
- Pulsation damper requirement on HP discharge — depends on pump type (piston pumps typically need accumulator, multistage do not)

Two-Pass RO Membrane Train

KNOWN FAILURE MODES (4)

- Membrane fouling/scaling from inadequate pre-treatment causing pressure rise and flux decline
- FRP pressure vessel end-cap blow-out from improper retaining ring installation or fatigue cycling
- O-ring/brine-seal failure between elements causing internal bypass and salt passage spike
- Telescoping damage to membrane elements from pressure transients during startup/shutdown

OPEN QUESTIONS (3)

- Final element count and array configuration (6:1 vs 7:1 vessel:element) pending detailed projection (e.g. WAVE/IMSDesign) for site-specific TDS and temperature
- Whether second pass requires partial-flow (split-permeate) configuration to meet boron limits in warm tropical seawater
- Vessel manufacturer selection (Codeline vs Bel vs Protec) based on lead time and side-port vs end-port preference

Post-treatment Module

KNOWN FAILURE MODES (4)

- Calcite media exhaustion/channelling causing under-dosed hardness, low pH product water, and corrosion of downstream metallic fittings
- NaOCl dosing pump failure or hypochlorite degradation (loses ~10% strength per month at 25°C) leading to insufficient chlorine residual and microbiological breakthrough
- CO₂ over-injection causing low pH and excessive calcium dissolution, or under-injection causing calcite bed to remain undissolved (no remineralisation)
- pH/chlorine sensor fouling or drift producing false telemetry, masking off-spec product water delivery

OPEN QUESTIONS (3)

- Final calcite consumption rate and refill interval — depends on permeate CO₂ dose and target hardness, affecting site-visit OPEX
- Whether single-stage calcite is sufficient or a downstream NaOH trim is needed to reach pH 8.0+ across full flow range
- Local regulatory variation: some jurisdictions mandate fluoridation or alternative disinfection (chloramine, UV) which would change the dosing skid configuration

Clean-In-Place System

KNOWN FAILURE MODES (4)

- Chemical attack on pump seals or tank fittings due to incorrect chemical concentration or use of incompatible elastomers, causing leaks of corrosive fluid into the container
- Inadequate flush of residual cleaning chemicals before returning to service, contaminating product water and breaching NSF/ANSI 61 compliance
- Heater element scaling or dry-fire failure if level switch interlock fails, leading to tank damage or fire risk
- Cross-contamination between acid and alkaline cycles if tank is not properly rinsed, neutralising chemicals and rendering cleaning ineffective

OPEN QUESTIONS (3)

- Whether CIP tank volume should be sized for single-bank or whole-train cleaning, which affects tank footprint and water/chemical inventory
- Whether to specify automated dosing pumps and valve actuation (higher CAPEX, safer for unskilled operators) versus fully manual operation (lower cost, suits trained crew)
- Compatibility of selected EPDM/Viton elastomers with the full range of proprietary cleaning chemicals likely to be used across the unit's deployment scenarios

Control Panel & Motor Control Centre

KNOWN FAILURE MODES (4)

- VFD overheating or derating in hot container interior (>45 °C ambient) leading to nuisance trips during summer deployments
- Ingress of salt-laden humid air into MCC causing corrosion of busbars and PCB contamination
- PLC I/O channel failure or sensor drift causing false interlock trips and unplanned shutdowns
- Inrush current on HP pump start tripping undersized site genset or mains breaker

OPEN QUESTIONS (3)

- Final HP pump motor rating (45–95 kW range) — drives VFD frame size and incoming breaker rating
- Whether customer requires SCADA/remote telemetry as standard or option, affecting gateway and cybersecurity scope
- Container internal ambient temperature profile — may require active MCC cooling (vortex cooler or AC unit) rather than passive ventilation

7. Supplier shortlist (32)

Shortlist built by scoring each supplier in the directory (651 companies) and marketplace listings (28,315 listings) against each module's declared process + material. A low shortlist count usually means the directory doesn't yet have coverage for the project's niche — not that no match exists globally.

isoclad.co.uk

Isoclad

<https://www.isoclad.co.uk/company-overview/>

Contact: info@isoclad.co.uk

HQ not declared · ramp role unassigned · match score 26.5

Matched across 1 module: container_enclosure

SCORE BREAKDOWN

total	26.5
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.1
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Insulated Panels & Cladding Systems

carlroth.com

Carl ROTH

https://www.carlroth.com/de/de/filtration-separation/membranfilter/c/web_folder_399096

Contact: bestellungen@carlroth.de

HQ not declared · ramp role unassigned · match score 24.4

Matched across 1 module: ro_membrane_array

SCORE BREAKDOWN

total	24.4
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	21
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Laboratory Consumables & Filtration

fairfields.co.uk

Fairfield Control Systems

<https://www.fairfields.co.uk/fcs/solutions/industrial-automation/control-panels/>

Contact: sales@fairfields.co.uk

HQ not declared · ramp role unassigned · match score 33.1

Matched across 1 module: electrical_distribution

SCORE BREAKDOWN

total	33.1
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	22.7
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Industry: Energy
- Products

sulzer.com

Sulzer

<https://www.sulzer.com/de-ch/products/separation-technology>

HQ not declared · ramp role unassigned · match score 25.9

Matched across 1 module: Two-Pass RO Membrane Train

SCORE BREAKDOWN

total	25.9
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	22.5
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Separation Technology & Process Equipment

deltafilter.com

DELTAFILTER Filtrationssysteme GmbH

<https://deltafilter.com/>

Contact: info@deltafilter.com

HQ not declared · ramp role unassigned · match score 26.4

Matched across 1 module: intake_pretreatment

SCORE BREAKDOWN

total	26.4
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Filtration Systems

fluidsystem.com

Fluid System

<https://fluidsystem.com/en/company/>

Contact: info@fluidsystem.com

HQ not declared · ramp role unassigned · match score 26.3

Matched across 1 module: High-Pressure Pump & Energy Recovery Device

SCORE BREAKDOWN

total	26.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	22.9
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Hydraulic Systems

phoenixcontrol.co.uk

Phoenix Control Systems Ltd

<https://phoenixcontrol.co.uk>

Contact: sales@phoenixcontrol.co.uk

HQ not declared · ramp role unassigned · match score 33.4

Matched across 1 module: Control Panel & Motor Control Centre

SCORE BREAKDOWN

total	33.4
keyword	0
process	0
quality	0.4
industry	5
material	0
semantic	25
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Industry: Automotive
- Manufacturer

cube-glass.com

Cube Glass

<https://cube-glass.com/>

Contact: sales@cube-glass.com

HQ not declared · ramp role unassigned · match score 30.2

Matched across 1 module: container_enclosure

SCORE BREAKDOWN

total	30.2
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	19.8
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Industry: Automotive
- Products

waterhydraulics.co.uk

The Water Hydraulics Company Ltd

<https://www.waterhydraulics.co.uk/>

Contact: sales@waterhydraulics.co.uk

HQ not declared · ramp role unassigned · match score 27.0

Matched across 1 module: High-Pressure Pump & Energy Recovery Device

SCORE BREAKDOWN

total	27
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.6
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Products

accupanel.com

DC DISTRIBUTION BOARD

<https://www.accupanel.com/dc-distribution-board/>

HQ not declared · ramp role unassigned · match score 28.6

Matched across 1 module: electrical_distribution

SCORE BREAKDOWN

total	28.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	25.2
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

dp.nl

DP-Pumps

<https://dp.nl/en/about-us/>

Contact: dp@dp.nl

HQ not declared · ramp role unassigned · match score 26.5

Matched across 1 module: High-Pressure Pump & Energy Recovery Device

SCORE BREAKDOWN

total	26.5
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.1
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Products

membrapure.de

membraPure

<https://membrapure.de/de/reinstwasseranlagen/>

Contact: sales@membrapure.de

HQ not declared · ramp role unassigned · match score 26.3

Matched across 2 modules: Two-Pass RO Membrane Train, Post-treatment Module

SCORE BREAKDOWN

total	26.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	25.9
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Water Treatment Systems

pretreat.co.uk

PreTreat Ltd

<https://pretreat.co.uk/speciality-chemicals-for-metal-pretreatment/>

Contact: sales@pretreat.co.uk

HQ not declared · ramp role unassigned · match score 30.4

Matched across 1 module: post_treatment

SCORE BREAKDOWN

total	30.4
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	20
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Industry: Aerospace
- Products

automationcontrol.wales

Automation and Control Systems

<https://www.automationcontrol.wales/>

Contact: sales@automationcontrol.wales

HQ not declared · ramp role unassigned · match score 33.5

Matched across 1 module: plc_control

SCORE BREAKDOWN

total	33.5
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	23.1
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Industry: Automotive
- Services

getfse.com

FSE

<https://getfse.com/control-panels>

HQ not declared · ramp role unassigned · match score 28.3

Matched across 1 module: electrical_distribution

SCORE BREAKDOWN

total	28.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	24.9
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Industrial Control Panels & Automation

getfse.com

FSE

<https://getfse.com/control-panels>

Contact: sales@getfse.com

HQ not declared · ramp role unassigned · match score 28.2

Matched across 1 module: ISO Container Shell & Skid Frame

SCORE BREAKDOWN

total	28.2
keyword	0
process	0
quality	0.4
industry	5
material	0
semantic	19.8
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Industry: Aerospace
- Products

modular-cleanroom.net

Modular Cleanroom

<https://www.modular-cleanroom.net/>

Contact: info@tca.group

HQ not declared · ramp role unassigned · match score 29.0

Matched across 1 module: ISO Container Shell & Skid Frame

SCORE BREAKDOWN

total	29
keyword	0
process	0
quality	0.4
industry	5
material	0
semantic	23.6
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Industry: Aerospace
- Products

krone-filter.com

Krone Filter Solutions GmbH

<https://www.krone-filter.com/>

Contact: filter@krone-filter.com

HQ not declared · ramp role unassigned · match score 26.7

Matched across 1 module: Pre-treatment Module

SCORE BREAKDOWN

total	26.7
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.3
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Industrial Filtration Systems

nammo.com

Nammo UK Ltd (Nammo Westcott)

<https://www.nammo.com/location/westcott/>

Contact: sales@nammo.com

HQ not declared · ramp role unassigned · match score 20.4

Matched across 1 module: Two-Pass RO Membrane Train

SCORE BREAKDOWN

total	20.4
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	0
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Industry: Aerospace
- Regulated-industry certs
- Hardware Company

prominent.com

ProMinent

<https://www.prominent.com/en/Products/Products/Membrane-Technology-and-Membrane-Filtration/pg-membrane-technology-and-filtration.html>

HQ not declared · ramp role unassigned · match score 27.6

Matched across 2 modules: Pre-treatment Module, Post-treatment Module

SCORE BREAKDOWN

total	27.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	24.2
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Membrane Filtration & Metering Systems

precisioncleanrooms.co.uk

Precision Cleanrooms

<http://www.precisioncleanrooms.co.uk/>

Contact: enquiries@precisioncleanrooms.co.uk

HQ not declared · ramp role unassigned · match score 25.0

Matched across 1 module: Clean-In-Place System

SCORE BREAKDOWN

total	25
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	21.6
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Services

cdrpumps.co.uk

CDR Pumps (UK) Ltd

<https://www.cdrpumps.co.uk/>

Contact: sales@cdrpumps.co.uk

HQ not declared · ramp role unassigned · match score 25.9

Matched across 1 module: cip_chemical_dosing

SCORE BREAKDOWN

total	25.9
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	25.5
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Products

ppcanda.com

PP Control & Automation

<https://www.ppcanda.com/control-automation/>

Contact: info@ppcanda.com

HQ not declared · ramp role unassigned · match score 33.3

Matched across 1 module: plc_control

SCORE BREAKDOWN

total	33.3
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	22.9
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Industry: Automotive
- Services

spxflow.com

Seital Separation

<https://www.spxflow.com/seital-separation/>

Contact: dortmund.sc@spxflow.com

HQ not declared · ramp role unassigned · match score 26.7

Matched across 1 module: intake_pretreatment

SCORE BREAKDOWN

total	26.7
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.3
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Products

southwest-stainless.co.uk

SWS Engineering Ltd (Southwest Stainless)

<https://southwest-stainless.co.uk>

Contact: sales@southwest-stainless.co.uk

HQ not declared · ramp role unassigned · match score 27.9

Matched across 1 module: ISO Container Shell & Skid Frame

SCORE BREAKDOWN

total	27.9
keyword	0
process	0
quality	0.4
industry	5
material	0
semantic	19.5
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Industry: Aerospace
- Manufacturer

inoxpassivation.co.uk

Inoxpassivation

<https://www.inoxpassivation.co.uk>

Contact: sales@inoxpassivation.com

HQ not declared · ramp role unassigned · match score 24.3

Matched across 1 module: Clean-In-Place System

SCORE BREAKDOWN

total	24.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	20.9
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Services

sm-control.co.uk

SM Control Engineering Ltd

<https://www.sm-control.co.uk>

Contact: sales@sm-control.co.uk

HQ not declared · ramp role unassigned · match score 35.2

Matched across 1 module: Control Panel & Motor Control Centre

SCORE BREAKDOWN

total	35.2
keyword	0
process	0
quality	0.4
industry	5
material	0
semantic	26.8
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Industry: Automotive
- Manufacturer

whgood.co.uk

WH Good

<https://www.whgood.co.uk/integrated-automation-process-solutions/control-panel-manufacture/>

Contact: automation@whgood.co.uk

HQ not declared · ramp role unassigned · match score 38.9

Matched across 1 module: Control Panel & Motor Control Centre

SCORE BREAKDOWN

total	38.9
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	25.5
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Industry: Automotive
- Products

filterdesigns.com

Filter Designs

<https://www.filterdesigns.com/>

Contact:

HQ not declared · ramp role unassigned · match score 27.3

Matched across 1 module: Pre-treatment Module

SCORE BREAKDOWN

total	27.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.9
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Services

mottcorp.com

Mott Corp

<https://www.mottcorp.com/de/product/process-filters-skids-elements/coalescers/>

Contact: sales@mottcorp.com

HQ not declared · ramp role unassigned · match score 41.4

Matched across 1 module: Two-Pass RO Membrane Train

SCORE BREAKDOWN

total	41.4
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	21
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Industry: Aerospace
- Regulated-industry certs
- Industrial Filtration & Fluid Processing

watchwater.de

Watch Water

<https://www.watchwater.de/>

Contact: info@watchwater.de

HQ not declared · ramp role unassigned · match score 27.0

Matched across 1 module: Post-treatment Module

SCORE BREAKDOWN

total	27
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	26.6
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Water Treatment Media & Chemicals

ipbchemicalcleaning.co.uk

IPB Chemical Cleaning

<https://ipbchemicalcleaning.co.uk/>

Contact: sales@ipbchemicalcleaning.co.uk

HQ not declared · ramp role unassigned · match score 25.3

Matched across 1 module: Clean-In-Place System

SCORE BREAKDOWN

total	25.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	24.9
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Services